

Deep Learning for Perception Course Outline

Week 1: Introduction and Transition

- Recap: supervised/unsupervised learning, overfitting, generalization
- Intro to deep learning and perception use cases
- Tools: Tensorflow, Google Colab, Kaggle setup

Week 2: Multilayer Perceptrons

- MLP structure, forward/backpropagation
- Activation functions (ReLU, sigmoid, tanh)
- Loss functions (MSE, BCE)
- Gradient descent and optimizers

Week 3: Introduction to CNNs

- Concept of local connectivity and shared weights
- Filters, convolution, stride, padding
- Pooling (max, avg)
- RGB image convolution walkthrough

Week 4: CNN Architecture and Feature Maps

- Full CNN pipeline
- Feature map size calculation
- Memory and parameter estimation
- Hands-on: LeNet and AlexNet walkthrough
- <https://www.coursera.org/projects/googlecloud-introduction-to-computer-vision-with-tensorflow-7q5na>

Week 5: Advanced Architectures

- VGG: simplicity and depth
- ResNet: residual blocks
- GoogleNet: inception modules
- MobileNet: depthwise separable convolutions
- Comparison of design tradeoffs

Week 6: Image Classification

- Output layer design (softmax, BCE)
- Evaluation metrics: accuracy, precision, recall
- Hands-on: train a classifier on CIFAR-10 or MNIST
- <https://www.coursera.org/projects/tensorflow-beginner-basic-image-classification>

Week 7: Object Detection

- Challenges in detection vs classification
- Sliding window vs region proposals
- YOLO framework: intuition, structure
- Hands-on: inference with pretrained YOLO
- <https://www.coursera.org/projects/object-localization-tensorflow>

Week 8: Image Segmentation

- Types: semantic vs instance segmentation
- U-Net architecture
- Hands-on: segmenting medical or road scene datasets
- <https://www.coursera.org/projects/googlecloud-introduction-to-computer-vision-with-tensorflow-7q5na>

Week 9: Transformers in Vision

- Why CNNs are limited
- Vision Transformer (ViT): patch embeddings, positional encoding
- Transformer block mechanics

Week 10: Temporal Perception

- Sequential data and perception
- RNNs, vanishing gradient issue
- LSTM: gates and flow
- Use case: action recognition or video captioning

Week 11: Generative Models

- Autoencoders: encoder-decoder pipeline
- Variational autoencoders (brief)
- GANs: architecture, discriminator vs generator
- GAN types: DCGAN, Pix2Pix, CycleGAN
- Applications: image translation, super-resolution

Week 12: Perception with Multimodal Data

- Combining text and vision
- CLIP: contrastive pretraining
- ALIGN: large-scale vision-language alignment
- Hands-on: zero-shot classification using CLIP

Week 13–14: Project Work

- Project building and checkpoint reviews
- Students apply methods to novel perception tasks

Week 15: Final Presentations

- Project demonstrations
- Peer review and discussion

Let me know if you want slides, datasets, or lab ideas per week.